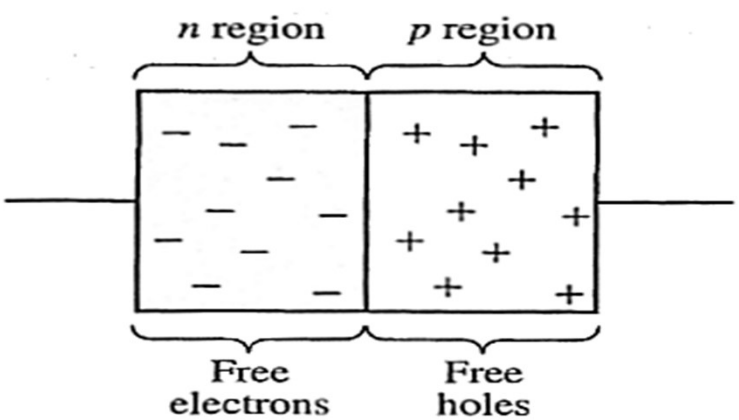
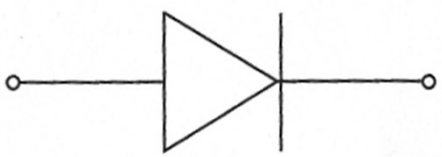
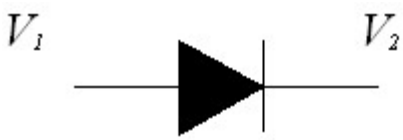
# Diodes

It is represented by the following symbol, where the arrow indicates the direction of positive current flow.

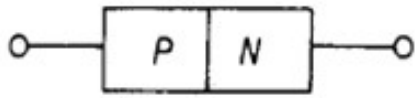


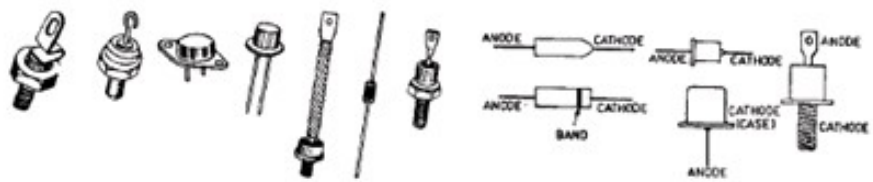



# P-N Junction



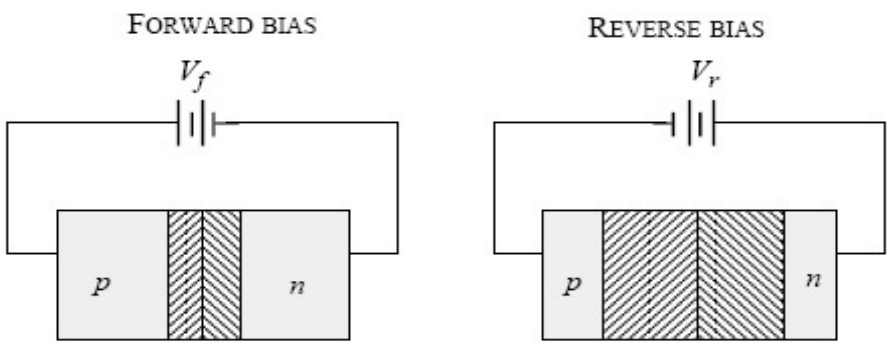

- $V_1 > V_2$, current →
- $V_1 < V_2$, no current



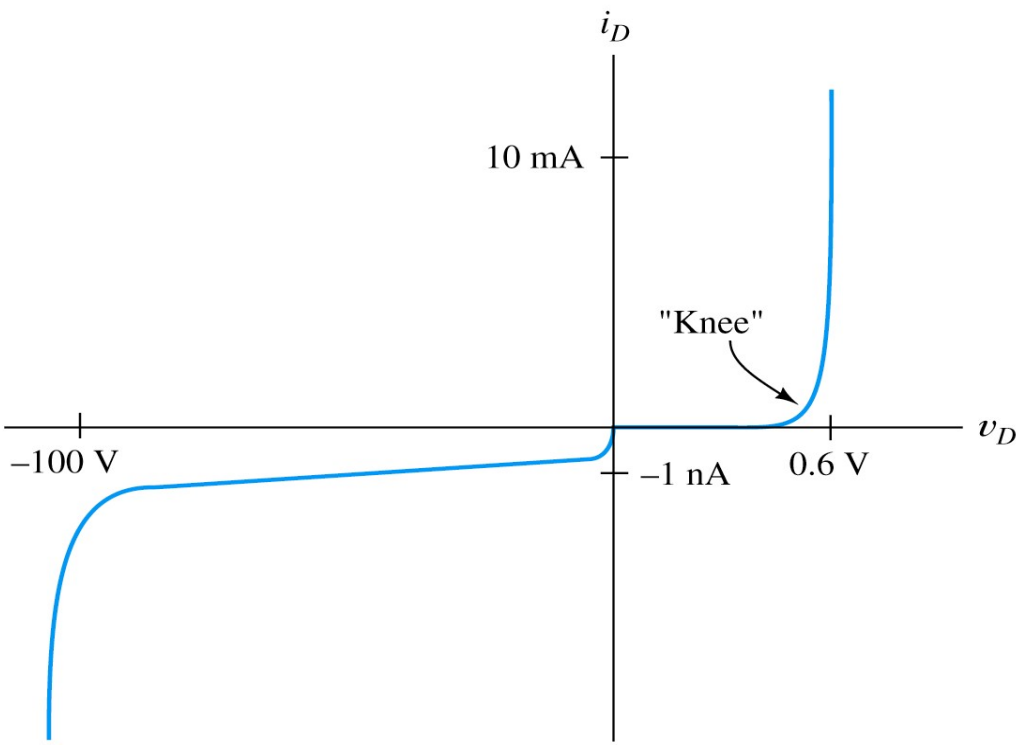


# Forward Bias and Reverse Bias

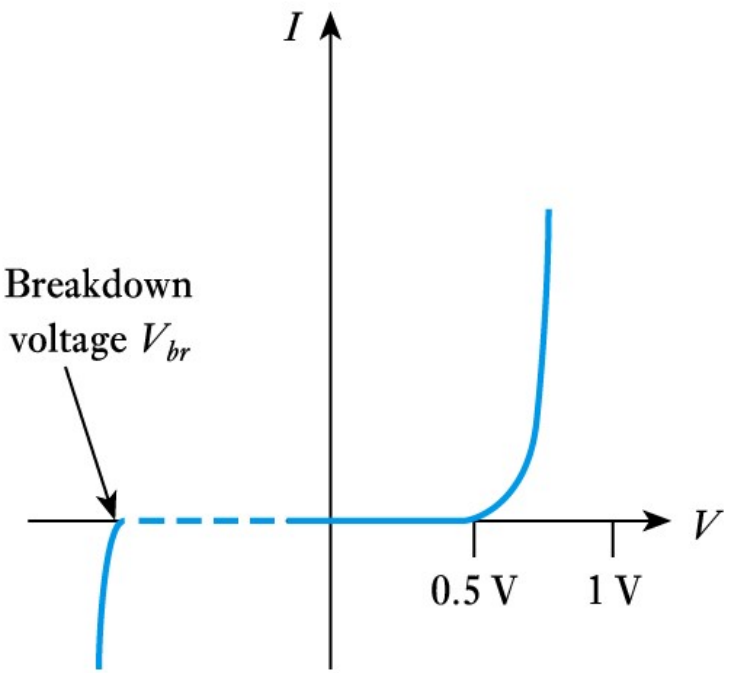
- Forward Bias : Connect positive of the Diode to positive of supply…negative of Diode to negative of supply
- Reverse Bias: Connect positive of the Diode to negative of supply…negative of diode to positive of supply.



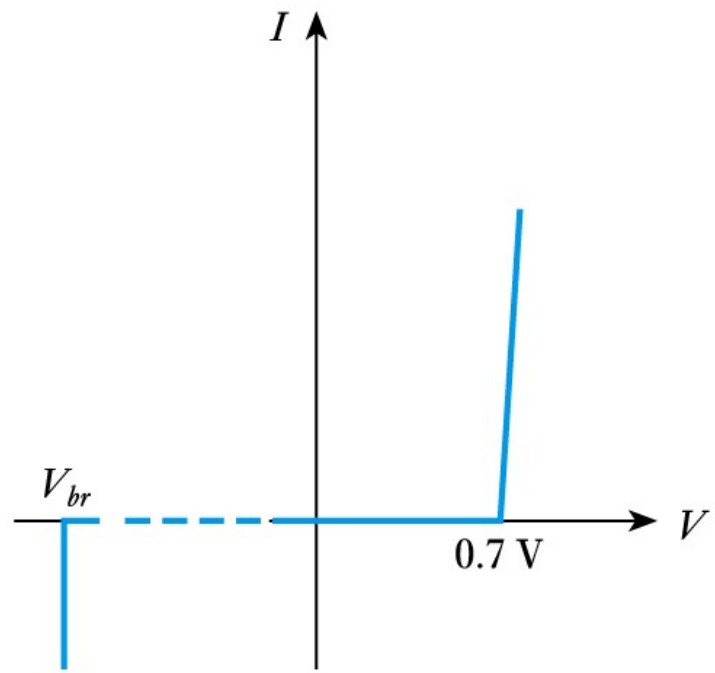
# I-V Characteristics of  Practical Diode



- **Turn-on** and **breakdown voltages** for a silicon




(a) A silicon diode




(b) Straight-line approximation to silicon diode characteristics

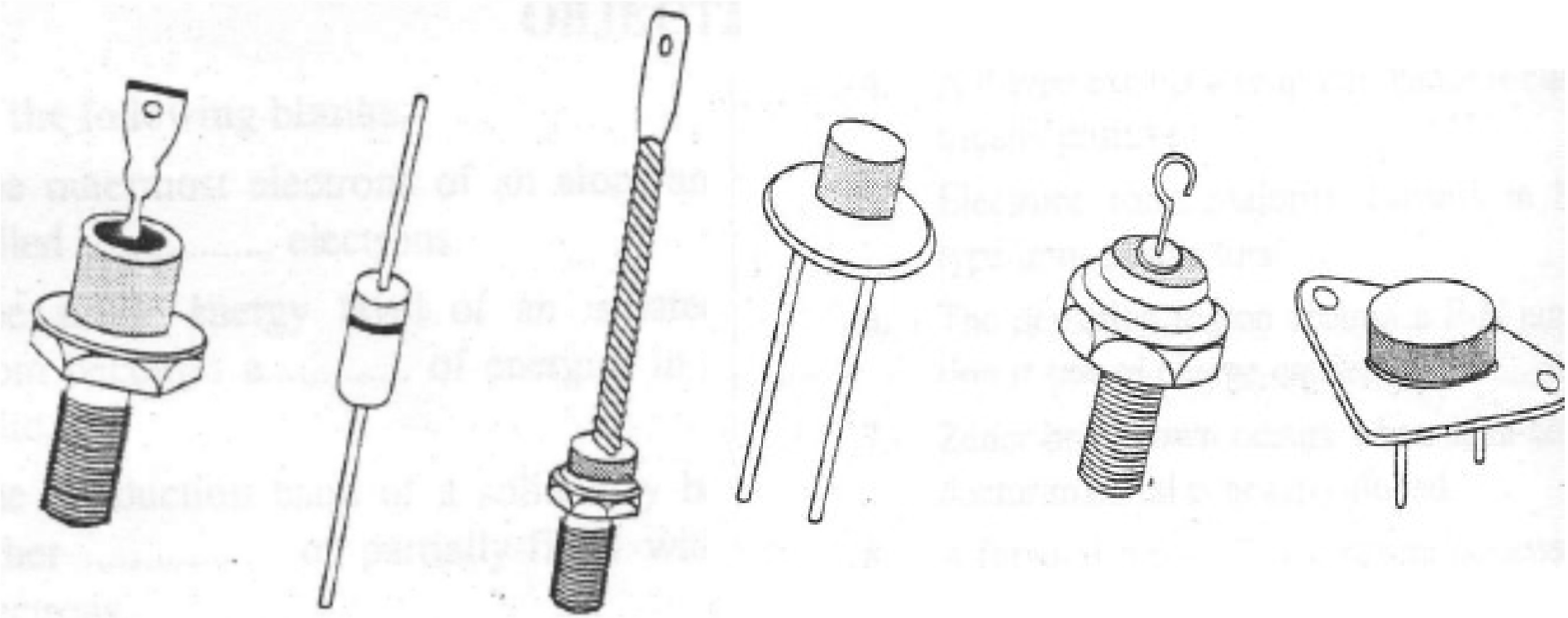
# Shockley Equation

$$i_D = I_s\left[\exp\left(\frac{v_D}{nV_T}\right) - 1\right] \qquad V_T = \frac{kT}{q}$$

$$V_T \cong 26\,\mathrm{mV}$$

$I_s$ is the saturation current ~$10^{-14}$
$V_d$ is the diode voltage
n – emission coefficient (varies from 1 - 2 )
$k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant
$q = 1.60 \times 10^{-19}$ C is the electrical charge of an electron.
At a temperature of 300 K, we have

# Physical structures



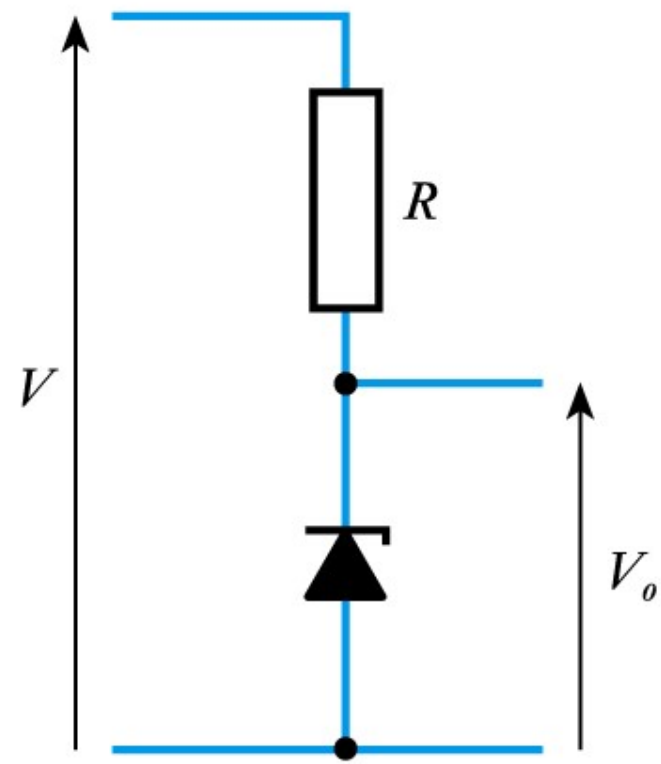
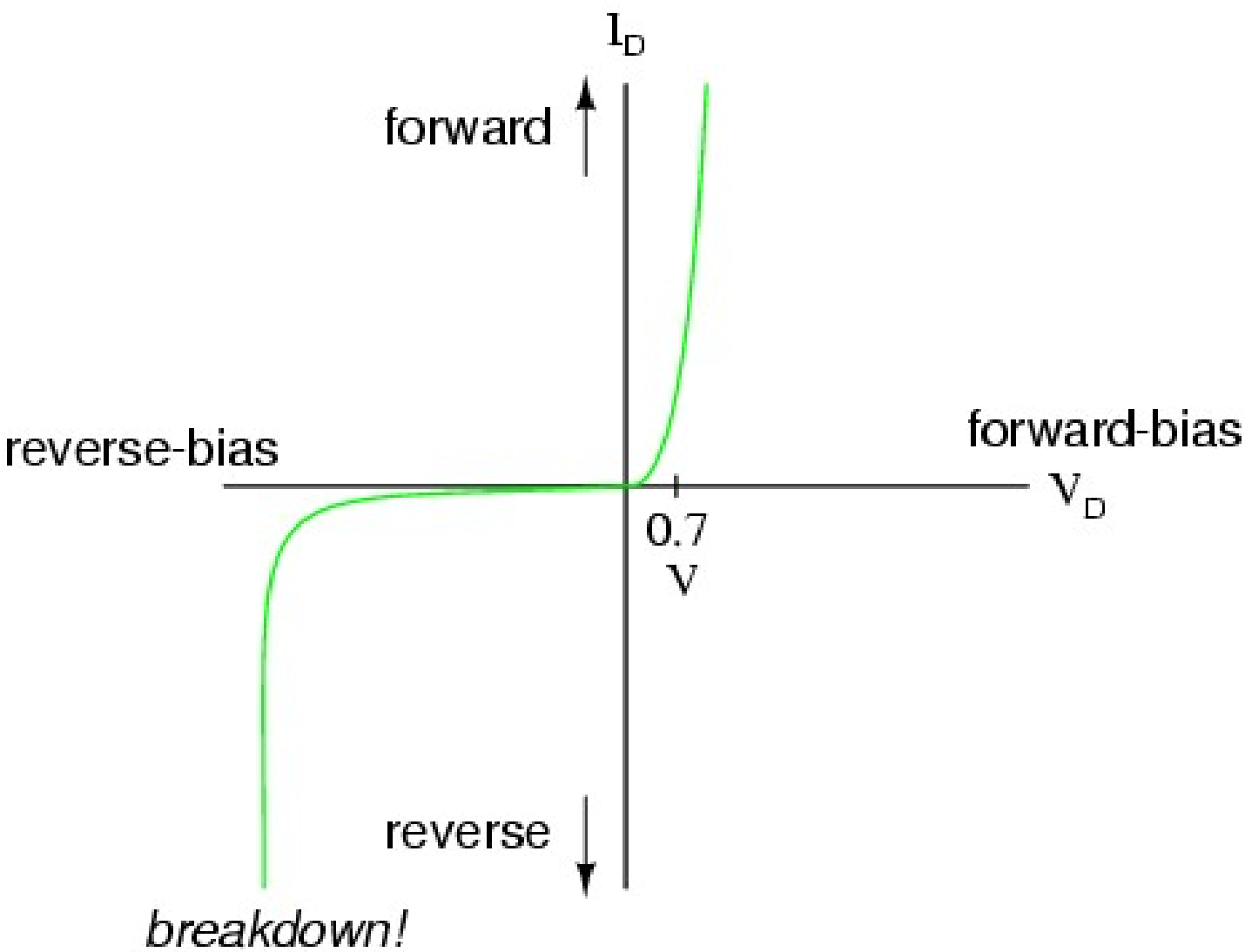
- **Silicon diodes**
  - generally have a turn-on voltage of about 0.5 V
  - generally have a conduction voltage of about 0.7 V
  - have a breakdown voltage that depends on their construction
    - perhaps 75 V for a **small-signal diode**
    - perhaps 400 V for a **power device**
  - have a maximum current that depends on their construction
    - perhaps 100 mA for a **small-signal diode**
    - perhaps many amps for a **power device**

- **Schottky diodes**
  - formed by the junction between a layer of metal (e.g. aluminium) and a semiconductor
  - action relies only on majority charge carriers
  - much faster in operation than a *pn* junction diode
  - has a low forward voltage drop of about 0.25 V
  - used in the design of high-speed logic gates

# Zener diodes

- A Zener diode is a pn silicon junction diode that is reverse biased and with the supply voltage sufficient to produce the ‘avalanche’ or ‘breakdown’ effect.
- A zener diode must be used in conjunction with a ‘current-limiting’ resistor to keep the current flow through the diode to a safe level.
- A zener diode is used to produce a stable voltage from a supply which is fluctuating.

- **Zener diodes**
  - uses the relatively constant reverse breakdown voltage to produce a voltage reference
  - breakdown voltage is called the **Zener voltage**, $V_Z$
  - output voltage of circuit shown is equal to $V_Z$ despite variations in input voltage $V$
  - a resistor is used to limit the current in the diode



# V/I CHARACTERISTICS